

University of Waterloo
Faculty of Engineering
Department of Electrical and Computer Engineering

SixthSense: A Plug & Play Sensor Network

Project Specifications and Risk Assessment

Group 2018.009

Prepared by

[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

Consultant: Derek Wright

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1 High-Level Project Description

This section provides an overview of the SixthSense project. It includes reasons for choosing the project, the objective, and a description of the design of the system.

1.1 Motivation

The market for industrial sensor networks is expected to grow by over 500% within the next five years [1]. Industrial wireless sensor networks are a promising technology due to the flexibility and scalability that they provide [2]. Sensor networks have a wide range of potential applications outside of an industrial setting, making sensor networks a rapidly growing multi-billion-dollar market [3]. However, developing a wireless sensor network requires configuration and low-level integration [4] that a non-expert engineer may not have the expertise to complete. Time may also be spent debugging performance issues during setup [5]. A more accessible sensor networking solution would allow non-expert engineers to capitalize on this growing market without incurring these configuration, integration, and debugging costs.

1.2 Project Objective

The objective of the SixthSense project is to design a low-cost, plug-and-play sensor network for the non-expert engineering designer market that is deployed with minimal configuration time. It allows designers to concentrate on using the sensor data to prototype instead of diverting time towards the secondary task of configuring the sensor infrastructure.

1.3 Block Diagram

The proposed solution is a suite of sensor nodes each containing a different sensor that wirelessly communicates to a central hub, which aggregates the sensor data. The system architecture is broken down into three main subsystems: sensor nodes, a central hub, and software as shown in Figure 1 on the following page. Each of these subsystems and their respective components are described in sections 1.3.1-1.3.3.

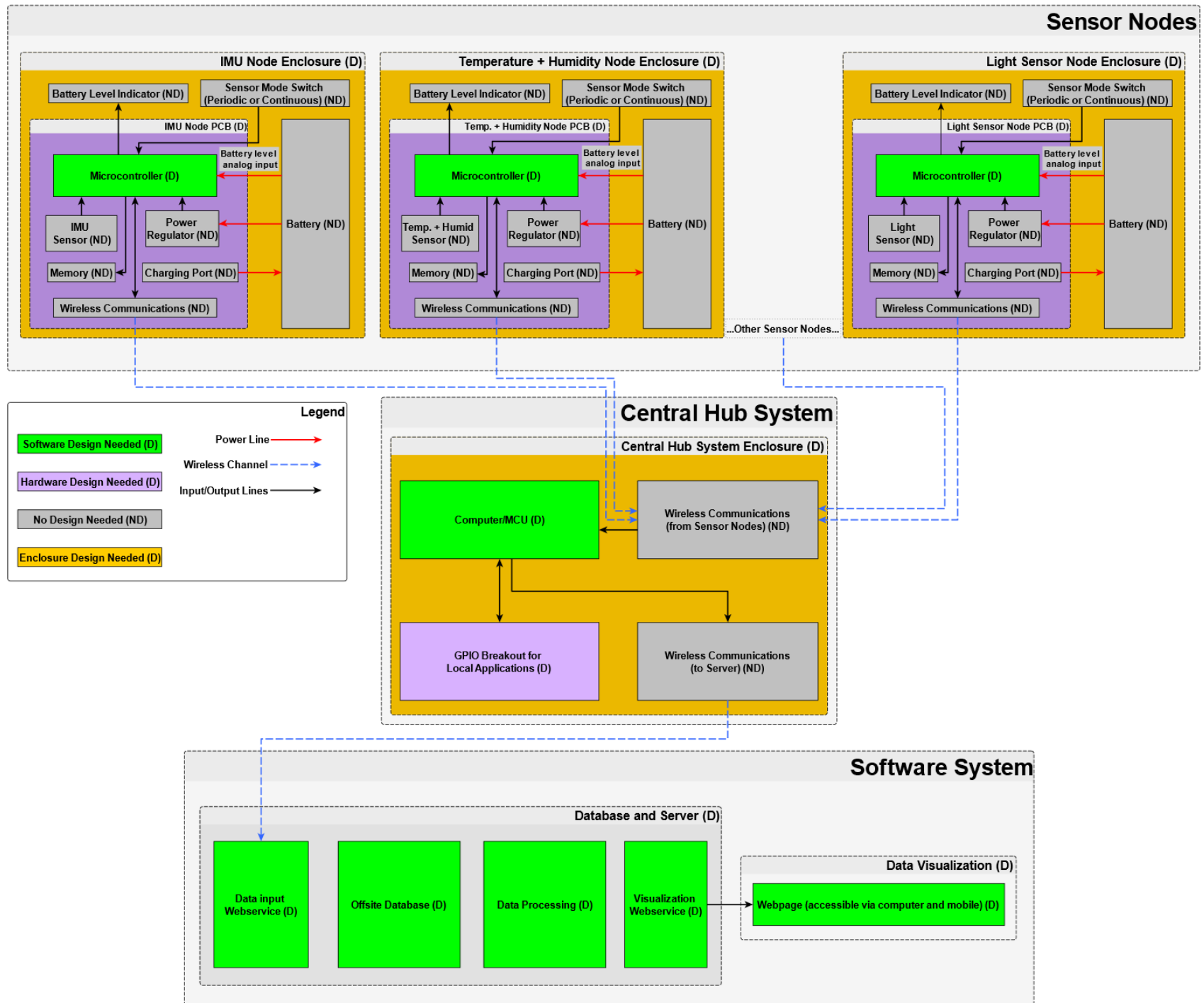


Figure 1. Block diagram of the SixthSense project

1.3.1 Sensor Nodes

This subsystem consists of the different sensor nodes that SixthSense offers. Each node consists of an enclosure, a custom-designed PCB, a battery, a battery level indicator, and a sensor mode switch. Each sensor node PCB has a similar architecture, with changes made to account for the interfacing requirements of different sensors. The battery level indicator displays the battery's charge state to the user. The user toggles the sensor mode switch between periodic sensor data retrieval and continuous sensor data retrieval depending on the application. The PCB contains power regulation hardware, a microcontroller to process the sensor data, memory to store sensor data onboard, and a wireless communication chip to transmit the sensor data to the central hub.

1.3.2 Central Hub System

The central hub system aggregates data from individual sensor nodes and stores it locally. The stored data is presented locally through the general-purpose input/output interface (GPIO) or by transmitting it to a server. The main component of the central hub is the microcontroller that runs software to collect and tabulate the sensor node data. The central hub also contains two wireless communications interfaces: one to collect data from individual sensor nodes and another to communicate with a server. The wireless interfaces may use different protocols.

1.3.3 Software System

The software system consists of the database with server and the data visualization subcomponent. The database is located offsite (on a cloud platform) and web services exist to retrieve the incoming data, process it, and display it to the user. Finally, the user can view sensor data and analytics on a webpage. Both the web services for data processing and the webpage for data visualization constitute the data visualization subcomponent.

2 Project Specifications

The project specifications are classified as either functional or non-functional specifications. The functional specifications describe what the system does while the non-functional specifications describe characteristics of the system. Each specification is further sub-categorized as essential or non-essential. Essential specifications are those that must be met for the design to be satisfactory. Non-essential specifications make the final product more than satisfactory when they are met. They are non-critical aspects of the system, and the system will still function properly if they are not met. Each specification is rated as essential or non-essential based on importance to the overall design.

2.1 Functional Specifications

Table 1 below outlines eight essential functional specifications for the SixthSense solution to be satisfactory. Table 2 follows listing three non-essential functional specifications.

Table 1. Essential functional specifications.

Specification	Subsystem	Description
Communication	Sensor Nodes	Each independent sensor node must be able to send data wirelessly to the central hub
Sensor network range	Sensor Nodes, Central Hub System	Sensor nodes must be able to connect to a central hub that is within a 30 m radius
Latency	Sensor Nodes, Central Hub System	The maximum latency (time taken to traverse wireless channel) between any sensor node and the central hub must be no more than 250 ms when 5 sensor nodes are simultaneously connected to the central hub and continuously recording data
Sensor integration	Sensor Nodes	The sensor network must be able to support both analog and digital sensors
Battery life during periodic monitoring	Sensor Nodes	Sensor nodes should be able to operate for 12 hours when sensors are set to acquire sensor data and log it by sending to the central hub periodically (every 5 minutes)
Battery life during continuous monitoring	Sensor Nodes	Sensor nodes should be able to operate for 1 hour when sensors are set to acquire sensor data and log it by sending to the central hub continuously
Rechargeability	Sensor Nodes	The sensor node battery must be able to recharge to full capacity within 2 hours
Data visualization	Software System	Must provide a web interface through which the user can view the data gathered by each of the sensor nodes that are connected to the central hub

Table 2. Non-essential functional specifications.

Specification	Subsystem	Description
Smartphone application	Software System	A smartphone application must be able to receive and display data gathered by each of the sensor nodes
Sensor disconnect notification	Sensor Nodes, Software System	The web interface must be able to detect and notify the user when a sensor node disconnects from the central hub
Data storage	Software System	The system must be able to store a minimum of 8 GB of total sensor data at any time in the database

2.2 Non-functional Specifications

The three essential non-functional specifications describing essential characteristics of the SixthSense project are described in Table 3.

Table 3. Essential non-functional specifications.

Specification	Subsystem	Description
Dimensions for sensor node PCB	Sensor Nodes	The length and width of a sensor node PCB must not exceed 7 cm x 4 cm
Support for different sensors	Sensor Nodes	SixthSense must be able to support a minimum of three distinct types of sensors
Battery level indicator	Sensor Nodes, Software System	The web interface and sensor node PCB must be able to indicate the charge remaining in the sensor node

The three non-essential non-functional specifications for the SixthSense design are outlined in Table 4.

Table 4. Non-essential non-functional specifications.

Specification	Subsystem	Description
Dimensions for central hub	Central Hub System	The central hub must fit within a 14 cm x 10 cm x 10 cm enclosure
Sensor setup time	Overall System	A user must be able to connect a new sensor node to a preconfigured central hub and start seeing data on the web interface in less than 5 minutes
Sensor node cost	Sensor Nodes	The cost of all the parts, including PCB fabrication and enclosure and excluding the actual sensor (due to variability in cost) must not exceed \$75.00 CAD

3 Risk Assessment

Table 5 below summarizes the most significant situations that could prevent successful completion of the project as well as strategies to mitigate each potential risk.

Table 5 Risk assessment and strategies to mitigate risks associated with the SixthSense project.

Potential Risk	Impact	Probability	Mitigation
Fabrication time for PCBs and enclosures – The layout and fabrication of PCBs as well as the design and manufacturing of the enclosures is an iterative and time-consuming process. Since this project has a very limited timeframe, any delays in the fabrication of these PCBs and enclosures can lead to incompleteness of the project.	Medium	High	To reduce the likelihood of this risk, vendors with minimal lead times on fabrication will be chosen. Stop dates on design will be in advance of final deadlines to allow minimum goals to be met. The group’s advisor is also experienced in PCB design and has industry connections. In the event of lead time issues, off-the-shelf microcontrollers and sensors already integrated on a board will be used to eliminate the time to design a custom PCB. The negative impact of pre-made boards is that enclosure dimension specs may not be met.

Potential Risk	Impact	Probability	Mitigation
Location of group members – The group consists of five 4A students from Electrical and Computer Engineering. Each of the members will work at their final co-op position from September 2017 to December 2017. The location of the employers may be worldwide (not local to the Greater Toronto Area), so direct, in-person collaboration on the project during this time may not be possible.	Medium	High	A weekly schedule as well as strict deadlines for major deliverables throughout the 4A term will ensure that minimum work during the co-op term is required to successfully complete the project. Also, delegating specific tasks and allowing each member of the team to take responsibility of certain components within the system will allow progress to be made on the project without the need for team collaboration. The negative impact of group members being far from each other during the co-op term is that integration of different members' subsystems may not occur during the co-op term, leaving this integration work to be completed in the 4B term.
Parts procurement – The design of this project will require purchasing of many electrical and hardware components. These items will be ordered from and shipped by various local and international sources which means there are potential risks of damages and delays.	High	Medium	Priority will be placed on finding alternative parts, suppliers, and vendors during the prototype stage in 4A. When the PCBs are designed, the group will be aware of shipment times and there will be alternative vendors to fulfill requests when the preferred vendors are not able to fulfill orders. The negative impact of damaged or delayed parts means that off-the-shelf boards may be purchased and used for the project, so specifications for enclosure dimensions might not be met.

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